1. Entropy of a 4-symbol Source

Symbols:
s_0, s_1, s_2, s_3
Probabilities:
$p_0 = 1/3$, $p_1 = 1/6$, $p_2 = 1/4$, $p_3 = 1/4$

The entropy is:
$H = -\sum_{i=0}^{3} p_i \log_2 p_i$

Calculate each term:

- $p_0 \log_2 p_0 = \frac{1}{3}\log_2\left(\frac{1}{3}\right) = \frac{1}{3} \times (-1.585) = -0.528$
- $p_1 \log_2 p_1 = \frac{1}{6}\log_2\left(\frac{1}{6}\right) = \frac{1}{6} \times (-2.585) = -0.431$
- $p_2 \log_2 p_2 = \frac{1}{4}\log_2\left(\frac{1}{4}\right) = \frac{1}{4} \times (-2) = -0.5$
- $p_3 \log_2 p_3 = \frac{1}{4}\log_2\left(\frac{1}{4}\right) = -0.5$

Sum:

$$H = -(-0.528 - 0.431 - 0.5 - 0.5) = 0.528 + 0.431 + 0.5 + 0.5 = \boxed{1.959 \text{ bits}}$$

2. Entropy of a Single Roll of a Fair Die

A fair die has 6 outcomes: each with probability 1/6.

$$H = -\sum_{i=1}^{6} \frac{1}{6} \log_2 \frac{1}{6}$$

$$\log_2\left(\frac{1}{6}\right) = -2.585$$

$$H = 6 \times \frac{1}{6} \times 2.585 = \boxed{2.585 \text{ bits}}$$

3. Plot Information vs Probability

The information gained by the occurrence:

$$I(p) = -\log_2 p$$

- At $p = 1$, $I(1) = 0$ bits (no surprise)
- As $p \to 0$, $I(p)$ grows large

A rough plot: **start at 0 (for p=1), rises slowly for large p, and rises sharply for small p**

UIT2521 – Information Theory and Applications

UNIT II INFORMATION THEORY FUNDAMENTALS

Tutorial – II

Entropy

Date & Time: 18.07.2025 (Friday, 1st & 2nd Hours)

1. A source emits one of four symbols s_0, s_1, s_2, s_3 with probabilities 1/3, 1/6, 1/4 and 1/4 respectively. The successive symbols emitted by the source are statistically independent. Calculate the entropy of the source.

2. Let X represents the outcome of a single roll of a fair die. Determine the entropy of X.

3. Let p denotes the probability of some event. Plot the amount of information gained by the occurrence of this event for $0 \le p \le 1$.

4. a) Calculate the average information in bits/character for the English language, assuming that, each of the 26 characters in the alphabet occurs with equal likelihood. Neglect spaces and punctuation.

   b) Since the alphabetic characters do not appear with equal frequency in the English language (or any other language), the answer to part a) will represent an upper bound on average information content per character. Repeat part a) under the assumption that the alphabet characters occur with the following probabilities:

p= 0.10 :	for letters	a, e, o, t
p= 0.07 :	for letters	h, i, n, r, s
p= 0.02 :	for letters	c, d, f, l, m, p, u, y
p= 0.01 :	for letters	b, g, j, k, q, v, w, x, z

5. Consider a discrete memoryless source with source alphabet S= {s_0, s_1, s_2}, and source statistics {0.7, 0.15, 0.15}. Calculate the entropy of the source.

4a. Avg Info (Entropy) for Uniform English Alphabets

26 letters, each with probability $1/26$:

$$H = -\sum_{i=1}^{26} \frac{1}{26} \log_2 \frac{1}{26} = -26 \times \frac{1}{26} \log_2 \frac{1}{26} = \log_2(26)$$

$$\log_2(26) \approx 4.700$$

So, **average information** = $\boxed{4.7 \text{ bits/character}}$

4b. English Alphabet, Nonuniform Frequencies

Probabilities:

- p=0.10: 4 letters (a, e, o, t)
- p=0.07: 5 letters (h, i, n, r, s)
- p=0.02: 8 letters (c, d, f, l, m, p, u, y)
- p=0.01: 9 letters (b, g, j, k, q, v, w, x, z)

Calculate entropy:

Calculate entropy:

$$H = -\sum_i p_i \log_2 p_i$$

$$H = -[4 \times 0.10 \log_2 0.10 + 5 \times 0.07 \log_2 0.07 + 8 \times 0.02 \log_2 0.02 + 9 \times 0.01 \log_2 0.01]$$

Plug in values:

- $\log_2(0.10) = -3.322$
- $\log_2(0.07) \approx -3.836$
- $\log_2(0.02) = -5.644$
- $\log_2(0.01) = -6.644$

So,

$$\begin{aligned} H = -[&4 \times 0.10 \times -3.322 \\ &+5 \times 0.07 \times -3.836 \\ &+8 \times 0.02 \times -5.644 \\ &+9 \times 0.01 \times -6.644] \end{aligned}$$

Compute each term:

- $4 \times 0.10 \times 3.322 = 1.329$
- $5 \times 0.07 \times 3.836 = 1.341$
- $8 \times 0.02 \times 5.644 = 0.903$
- $9 \times 0.01 \times 6.644 = 0.598$

Sum: $1.329 + 1.341 + 0.903 + 0.598 = 4.171$

So, **average information** = $\boxed{4.17 \text{ bits/character}}$

5. Entropy for Source: S = {s_0, s_1, s_2} with Probabilities {0.7, 0.15, 0.15}

$$H = -\sum_i p_i \log_2 p_i = -[0.7 \log_2 0.7 + 0.15 \log_2 0.15 + 0.15 \log_2 0.15]$$

- $\log_2(0.7) \approx -0.515$
- $\log_2(0.15) \approx -2.737$

Thus,

$$\begin{aligned} H = -[&0.7 \times -0.515 + 2 \times 0.15 \times -2.737] \\ &= [+0.3605 + 0.8211] \\ &= 0.3605 + 0.82 = \boxed{1.1816 \text{ bits}} \end{aligned}$$

Entropy formula:

$$H = -\sum_i p_i \log_2 p_i$$

Plug in the probabilities:

$$H = -[0.7 \log_2(0.7) + 0.15 \log_2(0.15) + 0.15 \log_2(0.15)]$$

First, calculate each term (using $\log_2$):

- $0.7 \log_2(0.7) \approx 0.7 \times (-0.515) = -0.361$
- Each $0.15 \log_2(0.15) \approx 0.15 \times (-2.737) = -0.411$

There are two 0.15 terms, so:

- $2 \times -0.411 = -0.822$

Sum all terms:

$$H = -[-0.361 + (-0.822)] = -[-1.183] = 1.183 \text{ bits}$$